

SageMath: Deeper Insights into Cybersecurity

In the previous article in this SageMath series (published in the January 2025 issue of *OSFY*), we concluded our discussion of classical encryption techniques and moved on to the exploration of modern cryptography by looking at symmetric-key cryptography. In this ninth article in the series, we will continue the focus on symmetric-key cryptography.



In the last article, we wrote a SageMath program to implement a simple symmetric-key cryptographic algorithm called DES using a Python package called 'PyCryptodome'. As promised, we will discuss the implementation of DES briefly in this article.

Most cryptography textbooks covering DES introduce a simplified version called the Simplified Data Encryption Standard (S-DES) to explain its working. S-DES follows the same fundamental principles as DES but operates on a smaller scale. It is a block cipher that encrypts 8-bit plaintext blocks using a 10-bit key, which is processed to generate two 8-bit subkeys for encryption and decryption. For a detailed analysis of S-DES, I recommend the classic textbook 'Cryptography and Network Security: Principles and Practice' by William Stallings. Let's first explore the concept of block ciphers before examining how DES functions as one.

Block ciphers

A block cipher is an encryption method that processes data in fixed-size blocks, transforming plaintext into ciphertext using a secret key. A block cipher encrypts a block of N bits at a time (for example, DES uses 64-bit blocks). Each block is encrypted independently but may be linked to other blocks depending on the mode in which the block cipher operates.

In addition to DES, other examples of block ciphers include AES (Advanced Encryption Standard), which operates on a 128-bit block of data with key sizes of 128, 192, or 256 bits; Blowfish, which operates on a 64-bit block of data with a variable-length key ranging from 32 to 448 bits; and Twofish, which operates on a 128-bit block of data with key sizes up to 256 bits.

Now, let us return to our discussion of DES. It was developed by IBM and adopted as a standard by the US government in 1977. DES takes a 64-bit key but uses a 56-bit secret key after removing 8 parity bits from the original 64-bit key. Both the sender and the receiver must have the same key to encrypt and decrypt data. As mentioned earlier, DES is a block cipher. It encrypts data in 64-bit blocks, meaning it processes data in chunks of 64 bits at a time. Each 64-bit block undergoes 16 rounds of encryption, using a combination of substitutions and permutations (transpositions). These transformations mix the data in a complex way. After 16 rounds, the scrambled output obtained is the ciphertext (encrypted data). To decrypt, the same steps are applied in reverse, using the same key. Notice that DES utilises both the classical encryption techniques—substitution and transposition—which we have previously discussed in this series. However, these operations are executed in a much more advanced and intricate manner.

Almost all block ciphers employ a technique called a Feistel network, which is a cryptographic structure used to construct block ciphers. This network was developed by Horst Feistel, a German-American cryptographer, and was first used in the Feistel cipher, a predecessor to DES. Its simple yet elegant design consists of multiple rounds of processing.